



Andrea Morales-Garzón

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WORK EXPERIENCE

 **UNIVERSITY OF GRANADA** – GRANADA, SPAIN

POSTDOCTORAL FELLOW IN COMPUTER SCIENCE – 17/07/2025 – CURRENT

Postdoctoral fellowship, funded by the University of Granada through a competitive call, focusing on large language models and graph neural networks.

 **UNIVERSITY OF COPENHAGEN** – COPENHAGEN, DENMARK

VISTING SCHOLAR – 01/04/2025 – 31/05/2025

Visiting scholar at the NLP Section of the Department of Computer Science at the University of Copenhagen (Denmark), under the supervision of Daniel Hershcovich.

 **UNIVERSITY OF COPENHAGEN** – COPENHAGEN, DENMARK

VISTING SCHOLAR – 01/06/2024 – 30/09/2024

Visiting scholar at the NLP Section of the Department of Computer Science at the University of Copenhagen (Denmark), under the supervision of Daniel Hershcovich.

 **UNIVERSITY OF JEAN MONNET (CAMPUS UNIVERSITY OF LYON)** – LYON, FRANCE

VISITING SCHOLAR – 31/05/2022 – 30/09/2022

Laboratoire Hubert Curien at the University of Jean Monet (Campus University of Lyon), under the supervision of Amaury Habrard

 **UNIVERSITY OF GRANADA** – GRANADA, SPAIN

PHD FELLOW IN COMPUTER SCIENCE – 15/02/2021 – 16/07/2025

PhD fellowship, funded by the Andalusian Government through a competitive call for young researchers, focusing on the intersection of Natural Language Processing (NLP), data heterogeneity, and massive data sources. Conducted research and developed algorithms applied to health and nutrition domains.

 **UNIVERSITY OF BERGEN** – BERGEN, NORWAY

GUEST VISITOR – 19/09/2022 – 23/09/2022

Visitor at the Department of Computer Science of the Bergen University and Media Futures.

 **UNIVERSITY OF GRANADA** – GRANADA, SPAIN

Address Avda. del Hospicio, s/n, 18010, Granada, Spain

TECHNICAL ASSISTANT IN THE INTERNATIONAL PROJECTS OFFICE – 01/09/2021 – 14/02/2022

Assistant on the promotion of European and International research projects.

 **GESTIÓN SALUD Y NUTRICION S.A.** – OVIEDO, SPAIN

RESEARCHER IN H2020 RESEARCH PROJECT STANCE4HEALTH – 14/02/2019 – 31/08/2021

Researcher in the Stance4Health Horizon 2020 Project

- Conducted research and contributed to build an intelligent nutritional assessment tool. Design and creation of a international database of foods. Intelligent integration of food sources and national supermarket databases.
- Collaborated with multidisciplinary teams to build a reliable and accurate compositional database of foods.

● **EDUCATION AND TRAINING**

01/12/2021 – 16/07/2024 Granada, Spain
PHD IN COMPUTER SCIENCE University of Granada

Website www.ugr.es

Granada, Spain
MASTER'S DEGREE IN DATA SCIENCE University of Granada

Website <https://masteres.ugr.es/datcom/> | **Level in EQF** EQF level 7

09/10/2020 Granada, Spain
MASTER'S DEGREE IN COMPUTER ENGINEERING University of Granada

- occupational techniques (making of standard breads, fancy breads, cakes and pastries)
- science applied to food and equipment (microbiology, biochemistry, hygiene)
- occupational technology (basic principles, hygiene and safety)
- knowledge of business and its economic, legal and social context

Website <https://masteres.ugr.es/ingenieria-informatica/> | **Final grade** Cum Laude | **Level in EQF** EQF level 7 |

Thesis Predictive Learning in Nutrition

27/06/2018 Granada, Spain
BACHELOR'S DEGREE IN COMPUTER SCIENCE University of Granada

Website www.ugr.es | **Level in EQF** EQF level 6

19/09/2023 – 12/01/2024 Granada, Spain
INTRODUCTION TO UNIVERSITY TEACHING COURSE 4TH EDITION (90 HOURS) University of Granada

● **LANGUAGE SKILLS**

Mother tongue(s): **SPANISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **SKILLS**

Microsoft Office | Social Media | Programming with Python, Node js, Ruby, Bash, React, Js | C / C++, Python, Java, HTML, SQL | Experience with PHP, SQL, HTML, CSS, JavaScript | Python (Tensorflow, Keras and Pytorch) | Deep Learning, | Natural language Processing

● **PUBLICATIONS**

2021-2025

- Jose A. Díaz-García; **Andrea Morales-Garzón**; Karel Gutiérrez-Batista; Maria J. Martin-Bautista. Optimising Text Classification in Social Networks via Deep Learning-Based Dimensionality Reduction. Electronics. 2025, 14(17), 3426. ISSN 2079-9292. <https://doi.org/10.3390/electronics14173426>.
- **Andrea Morales-Garzón**, Ana María Rojas-Carvajal, Roberto Morcillo-Jimenez, Maria J. Martin-Bautista, Karel Gutiérrez-Batista. Personalised healthy food text recommendations through fuzzy linguistic variables: A generative AI-based approach. Applied Soft Computing. Volume 178, 2025, 113234. ISSN 1568-4946. <https://doi.org/10.1016/j.asoc.2025.113234>.

- **Andrea Morales-Garzón**, Paola Santos Peinado, Roberto Morcillo-Jimenez, Karel Gutiérrez-Batista, María J. Martín-Bautista. Service-oriented multi-platform for food computing: A mobile application for recipe adaptation to nutrition behaviours (AI2Cuisine). *Expert Systems with Applications*. Volume 281, 2025, 127603. ISSN 0957-4174. <https://doi.org/10.1016/j.eswa.2025.127603>.
- **Andrea Morales-Garzón**; Karel Gutiérrez-Batista, María J. Martín-Bautista (in press). AdaptaFood: An intelligent system to adapt recipes to specialised diets and healthy lifestyles. *Multimedia Systems*. Accepted 08/01/2025.
- Bartolome Ortiz Viso; **Andrea Morales-Garzón**; María J. Martín-Bautista; Maria-Amparo Vila. Evolutionary Approach for Building, Exploring and Recommending Complex Items With Application in Nutritional Interventions. *IEEE Access*. 11, pp. 65891 - 65905. 2023. DOI: <https://doi.org/10.1109/ACCESS.2023.3290918>.
- **Andrea Morales-Garzón**; Karel Gutiérrez-Batista; María J Martín-Bautista. Link prediction in food heterogeneous graphs for personalised recipe recommendation based on user interactions and dietary restrictions. *Computing*. pp.1 - 23. Springer, 2023.
- **Andrea Morales Garzón**; Juan Gómez Romero; María José Martín Bautista. A Word Embedding-Based Method for Unsupervised Adaptation of Cooking Recipes. *IEEE Access*. 9, pp. 27389 - 27404. IEEE, 10/02/2021.

● CONFERENCES AND SEMINARS

Oral communications

- **Andrea Morales Garzón**; Karel Gutiérrez Batista; Roberto Morcillo Jimenez; María José Martín Bautista. "User-friendly health-conscious recipe adaptation system using fuzzy linguistic variables". *International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems*. July 2024.
- Roberto Morcillo Jimenez; **Andrea Morales Garzón**; Karel Gutiérrez Batista; María José Martín Bautista. "Fuzzy Logic and Science Sport: a linguistic approach to Isometric force manifestation". *International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems*. July 2024.
- Gracia Maria Sánchez Pérez; **Andrea Morales Garzón**; María José Martín Bautista; Juan Carlos Sierra. Data mining and artificial intelligence for analyzing texts related to sexuality on social networks. *National Conference of Early Career Researchers: Promoting Interdisciplinarity (JIFFI)*. June 2024.
- **Andrea Morales Garzón**; Karel Gutiérrez Batista; Roberto Morcillo Jimenez; María José Martín Bautista. "How Tasty Is This Dish? Studying User-Recipe Interactions with a Rating Prediction Algorithm and Graph Neural Networks". *International Conference on Flexible Query Answering Systems* (pp. 107-117). September 2023.
- **Andrea Morales Garzón**; Gracia Maria Sánchez Pérez; Juan Carlos Sierra; María José Martín Bautista. "The Promise of Query Answering Systems in Sexuality Studies: Current State, Challenges and Limitations.". *International Conference on Flexible Query Answering Systems* (pp. 107-117). September 2023.
- **Andrea Morales Garzón**; Juan Gómez Romero; María José Martín Bautista. "Contextual Sentence Embeddings for Obtaining Food Recipe Versions". *International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems*. July 2022.
- Daniel Hinojosa Nogueira; **Andrea Morales Garzón**; Alicia López Maldonado; Beatriz Navajas Porras; Silvia Pastoriza de la Cueva; José Ángel Rufián Henares. "Development and evaluation of a nutritional database of traditional recipes from different European countries involved in the STANCE4HEALT project". *III Conference of Food, Nutrition and Dietetics. Combining community and personalized nutrition: new challenges*. November 2020.
- Daniel Hinojosa Nogueira; Alicia López Maldonado; Bartolomé Ortiz Viso; **Andrea Morales Garzón**; Silvia Pastoriza de la Cueva; José Ángel Rufián Henares. "Comparación de un menú semanal con diferentes bases de datos de alimentos europeas en el marco del Proyecto STANCE4HEALTH". *II Conference on Epidemiology and Public Health: Researching in times of crisis*. November 2020.
- **Andrea Morales Garzón**; Juan Gómez Romero; María José Martín Bautista. "A word embedding model for mapping food composition databases using fuzzy logic". *International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems*. June 2020.

Poster communications

- **Andrea Morales-Garzón**; Oscar A. Rocha; Sara Benel Ramirez; Gabriel Tuco Casquino; Alberto Medina. Healthy cooking with large language models, supervised fine-tuning, and retrieval augmented generation. *Proceedings of the LatinX in AI (LXAI) Research Workshop at NAACL 2024*.
- **Andrea Morales Garzón**; Juan Gómez Romero; María José Martín Bautista. "Semantic-aware transformation of short texts using word embeddings: An application in the Food Computing domain". *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Student Research Workshop*. April 2021
- **Andrea Morales Garzón**; Juan Gómez Romero; María José Martín Bautista. "Deep learning and text processing in the nutrition field: An application for automatic adaptation of cooking recipes". *At Science, Technology, Engineering and Mathematics (STEM) Multidisciplinary Conference for scientific dissemination in Seville, Andalusia, Spain*. February 2023.
- **Andrea Morales Garzón**; Juan Gómez Romero; María José Martín Bautista. "Predictive analysis in Nutrition". *At I Forum on Equality in Research at the University of Granada, Spain*. February 2022. **Honor mention**.

Invited lectures

- **Andrea Morales Garzón**. "Applications of Artificial Intelligence in Nutrition" Faculty of Pharmacy, University of Granada. May 2024.
- **Andrea Morales Garzón**. "Natural Language Processing: From Theory to Practice". At DyploFEST 2024, in the School of Computer and Telecom Engineering, University of Granada. May 2024.

PROJECTS

Project participation and contribution

- BAG-INTEL - Intelligent system for improved efficiency and effectiveness of the customs control of passenger baggage from international flight arrivals. Type: Horizon Europe European Project. Duration: 01/09/2023 - 31/08/2026. Principal Investigator: Maria J. Martin Bautista. Role: Project team.
- FEDERAMED: Flexible federated learning with heterogeneous sources for the preservation of the security and data privacy: Health applications. Type: National Project Duration: 01/09/2022 - 31/08/2025. Principal Investigator: Maria J. Martin Bautista. Role: Project team.
- DESIRE: DEep learning for expert-driven applications in the Sexuality REsearch field. Type: Project funded by University of Granada (Spain). Duration: 01/01/2024 - 31/12/2024. Principal Investigator: **Andrea Morales Garzón**.
- Artificial Intelligence Techniques for Nutritional Guidance and Assessment. Type: Project funded by Research Center for Information and Communication Technologies (Spain). Duration: 08/05/2024 - 31/12/2024. Principal Investigator: **Andrea Morales Garzón**.
- HealthIA: Applications of Artificial Intelligence in Healthcare. Type: Dissemination Project funded by Research Center for Information and Communication Technologies (Spain). Duration: 21/10/2024 - 31/12/2024. Principal Investigator: **Andrea Morales Garzón**.
- DESINFOSCAN: Desinformation monitoring and its social impact using Artificial Intelligence: application to civil security. Type: National Project. Duration: 01/12/2022 - 30/11/2024. Principal Investigator: Maria J. Martin Bautista. Role: Project team.
- 2023 European Researchers' Night. Type: Horizon Europe European Project. Duration: 29/09/2023 - 29/09/2023. Role: Participant.
- 2022 European Researchers' Night. Type: Horizon Europe European Project. Duration: 30/09/2023 - 30/09/2023. Role: Participant.

RESEARCH FELLOWSHIPS

17/07/2025 – CURRENT

Postdoctoral fellowship from the Intern Research plan of the University of Granada

31/05/2022 – 30/09/2022

Incoming Mobility Scholarship for Short Research Stays at Lyon and Saint-Étienne (France)

Awarding entity: The Arqus Alliance, in collaboration with the Agence Nationale de la Recherche (ANR).
Amount of the grant: 2.000 €

15/02/2022 – 15/02/2026

Predocctoral fellowship by agents of the Andalusian Knowledge System

Competitive grants to finance the pre-doctoral hiring of research personnel in training by agents of the Andalusian Knowledge System

01/09/2019 – 30/06/2020

Research Initiation Grant for Students of Official Master's Degrees at the University of Granada

Funded by the University of Granada

HONOURS AND AWARDS

19/06/2024

SCIE-ZONTA-SNGULAR Award from the Spanish Scientific Society of Computer Science (Runner-up award) – Spanish Scientific Society of Computer Science

The awards aim to recognize the contributions of women pursuing doctoral studies in the field of computer science, encourage their entry into research careers, and create role models for future generations,

20/06/2024

Business Idea Award from the Vice-Rectorate for Entrepreneurship at the University of Granada – University of Granada

26/01/2024

Ateneo de Granada - Fundación Unicaja 2023 Award in the Scientific and Technical Research Category – Ateneo Granada - Unicaja Foundation 2023

Honorable Mention

18/03/2024

Finalist in the 3 Minute Thesis scientific competition – University of Granada

12/03/2021

Award for the best Master's thesis of the Master's Degree in Computer Engineering – University of Granada

12/03/2021

Award for the best academic record of the 2019/2020 graduating class of the Master's Degree in Computer Engineering – University of Granada

First prize in the II Edition of the scientific competition "The Enigma" of the Faculty of Sciences of the University of Granada – University of Granada

● **TEACHING ACTIVITIES**

CURRENT

Teaching activities at University of Granada (2022 - ongoing)

- **A.Y. 2022-2023:** Module "**TIC in Health Care and Research Methodology**", Bachelor's Degree in Nursing, 1st-year course. Delivered 13.3 hours of practical instruction. University of Granada.
- **A.Y. 2022-2023:** Module "**Fundamentals of Databases**", Bachelor's Degree in Computer Engineering, 2nd-year course. Delivered 45 hours of practical instruction. University of Granada.
- **A.Y. 2023-2024:** Module "**Databases**", Bachelor's Degree in Information and Documentation, 2nd-year course. Delivered 60 hours of practical instruction. University of Granada.
- **A.Y. 2024-2025:** Module "**Design and Development of Information Systems**", Bachelor's Degree in Computer Engineering, 3rd-year course. Planned to deliver 45 hours of practical instruction. University of Granada.

● **SCIENTIFIC COMMUNICATION**

Scientific Communication Talks

- "Applications of Artificial Intelligence in Healthcare" at Science Week 2022. University of Granada. November 2024.
- "Artificial Intelligence in the Kitchen: Food Computing for Healthy Decisions" at European Researchers Night 2023. University of Granada. September 2023.
- "Association Rules, Hidden Patterns, and Their Application in Medical Data Analysis" at European Researchers Night 2023. University of Granada. September 2023.
- "How does an Artificial Intelligence system work?" at European Researchers Night 2022. University of Granada. September 2022.
- "The wisdom of crowds: Are many people smarter than one expert?" at European Researchers Night 2022. University of Granada. September 2022.
- "Artificial Intelligence for automatic generation of poems in Spanish" at European Researchers Night 2022. University of Granada. September 2022.
- "Tell me what you eat and I'll tell you who you are: Artificial Intelligence in the nutrition field" at Science Week 2022. University of Granada. November 2022.